

ABE 721

Agricultural Systems Modelling and Simulation

Module 1: Advanced Computer Aided Design and Analysis

Prepared by:
Kadeghe Fue, PhD

Electronics and Precision Agriculture Laboratory (EPAL)
Sokoine University of Agriculture (SUA)
Morogoro, Tanzania

Contents

1 Advanced Computer Aided Design and Analysis **2**

1.1 Geometric Modelling Techniques 2

1.1.1 Definition and Core Principles 2

1.1.2 Types of Geometric Models 2

1.1.3 Applications in Agriculture 4

1.2 Systems Analysis Basics 4

1.2.1 Definition 4

1.2.2 System Architecture 4

1.2.3 Taxonomy of Systems 5

1.3 Application to Farm Machinery Scheduling Problem 5

1.3.1 What is Scheduling? 5

1.3.2 Key Factors in Scheduling 5

1.3.3 Modelling Approach 5

1.4 Design of Farm Machinery Using CAD 6

1.4.1 What is CAD? 6

1.4.2 CAD in Agricultural Engineering 6

1.4.3 Finite Element Analysis (FEA) 6

1.4.4 Advantages 7

Chapter 1

Advanced Computer Aided Design and Analysis

1.1 Geometric Modelling Techniques

1.1.1 Definition and Core Principles

Geometric modelling constitutes the mathematical representation of the shape, structure, and spatial properties of physical objects within a computational environment. In the context of agricultural engineering, this forms the absolute foundation of Computer-Aided Design (CAD), enabling the transition from conceptual schematics to highly precise digital prototypes.

The models exist fundamentally in 2D or 3D Cartesian space, utilizing coordinate geometry and vector mathematics to define boundaries, surfaces, and volumes. These representations are not merely visual; they contain critical topological data that facilitates downstream engineering processes such as finite element analysis (FEA), kinematic simulation, and computer-aided manufacturing (CAM).

1.1.2 Types of Geometric Models

The evolution of CAD has yielded three primary paradigms for geometric representation, each with specific computational requirements and application domains.

1. Wireframe Models

A wireframe model is the simplest mathematical representation, consisting entirely of points (vertices) connected by lines and curves (edges).

- **Mathematical Basis:** Defined by a set of coordinates $V = \{v_1, v_2, ..., v_n\}$ and a set of edges $E = \{e_1, e_2, ..., e_m\}$ connecting these vertices.
- **Advantages:** Extremely low computational overhead; fast processing.

- **Limitations:** Highly ambiguous; lacks surface definition, making it impossible to calculate mass properties or perform stress analysis.

2. Surface Models

Surface models construct the "skin" or outer boundary of an object. They utilize complex parametric mathematical formulations to represent free-form shapes.

- **Parametric Representation:** Commonly utilizes Bézier curves and Non-Uniform Rational B-Splines (NURBS). A Bézier curve of degree n is defined by:

$$P(t) = \sum_{i=0}^{n} \binom{n}{i} (1-t)^{n-i} t^i P_i \quad \text{for } 0 \leq t \leq 1 \tag{1.1}$$

  where P_i are the control points.

- **Applications:** Essential in agricultural aerodynamics and fluid dynamics, such as designing the sweeping curve of a plough moldboard to minimize soil resistance.

3. Solid Models

Solid modelling is the most robust and complete representation, defining the internal volume, outer boundary, and mass of an object.

- **Constructive Solid Geometry (CSG):** Uses Boolean operations (Union, Intersection, Difference) on basic geometric primitives.

- **Boundary Representation (B-Rep):** Defines a solid by its topological boundaries (Faces, Edges, Vertices) ensuring that the surface is completely closed.

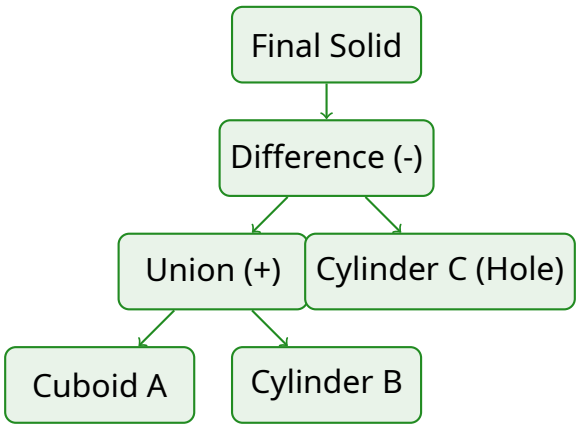


Figure 1.1: Constructive Solid Geometry (CSG) Boolean Tree Structure

1.1.3 Applications in Agriculture

In engineering contexts, geometric modelling is practically applied to:

- **Tillage Equipment Design:** Designing ploughs, harrows, and subsoilers.
- **Irrigation Systems:** Modelling layouts and components for precision water delivery.
- **Custom Machinery:** Prototyping specialized harvester components and machinery parts modelling.

1.2 Systems Analysis Basics

1.2.1 Definition

Systems analysis is the rigorous, methodical process of studying a complex entity to understand its constituent components, their intricate interactions, and the overall performance dynamics. A system is formally defined as a set of interconnected components working together toward a specific goal.

1.2.2 System Architecture

- **Inputs:** Resources entering the system boundary from the environment (e.g., solar radiation, precipitation, diesel fuel).
- **Processes:** The physical, biological, or chemical mechanisms transforming inputs.
- **Outputs:** The resulting yields or consequences (e.g., crop yield, machine wear, exhaust emissions).
- **Feedback:** Information regarding the output that is routed back to influence the input, crucial for system stability and control.

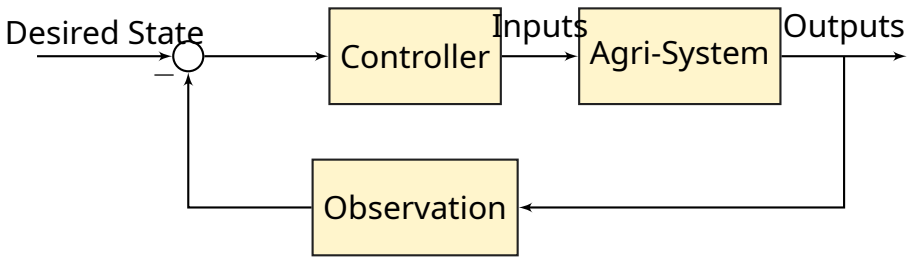


Figure 1.2: Standard Block Diagram of a Closed-Loop Agricultural System

1.2.3 Taxonomy of Systems

- **Deterministic vs. Stochastic:** Deterministic systems have predictable outcomes for given inputs. Agricultural systems are inherently *stochastic*; they involve random variables (weather patterns, disease outbreaks).
- **Static vs. Dynamic:** Static systems do not change over time. Dynamic systems evolve, requiring time-variant models.

1.3 Application to Farm Machinery Scheduling Problem

1.3.1 What is Scheduling?

Scheduling is the operational research process of allocating constrained resources (tractors, combine harvesters, operators, time) efficiently. Farm machinery must be scheduled to minimize delays, reduce operational costs, and maximize overall productivity.

1.3.2 Key Factors in Scheduling

- Availability of machines and implements.
- Weather conditions and soil states.
- Field size, shape, and geographical location.
- Labor availability.
- Biological crop timelines (planting windows, harvesting windows).

1.3.3 Modelling Approach

The scheduling problem requires a structured modelling approach:

1. **Define Objectives:** E.g., minimize total completion time or fuel consumption.
2. **Identify Constraints:** E.g., precedence (ploughing before planting), or limited tractor availability.
3. **Build a Model:** Construct mathematical or simulation models to represent the farm logic.
4. **Run Scenarios:** Use the model to optimize performance and test various strategies.

1.4 Design of Farm Machinery Using CAD

1.4.1 What is CAD?

Computer-Aided Design (CAD) is the use of computational software to design, analyze, simulate, and optimize products. It facilitates the creation of 2D drafting and complex 3D solid models, replacing manual drafting.

1.4.2 CAD in Agricultural Engineering

Modern CAD tools are utilized heavily in the design of equipment such as:

- Primary and secondary tillage implements (ploughs, harrows).
- Harvesting equipment.
- Precision irrigation systems.

Furthermore, CAD allows for the virtual testing of stress and durability (Computer-Aided Engineering or CAE) to improve efficiency before any physical production begins.

1.4.3 Finite Element Analysis (FEA)

Agricultural machinery is subjected to extreme, non-linear dynamic loads. FEA discretizes the continuous CAD solid model into a finite mesh of elements to calculate stress, strain, and deformation. The fundamental governing equation of static FEA is:

$$[K]\{U\} = \{F\} \tag{1.2}$$

Where $[K]$ is the Stiffness Matrix, $\{U\}$ is the Nodal Displacement Vector, and $\{F\}$ is the Applied Force Vector.

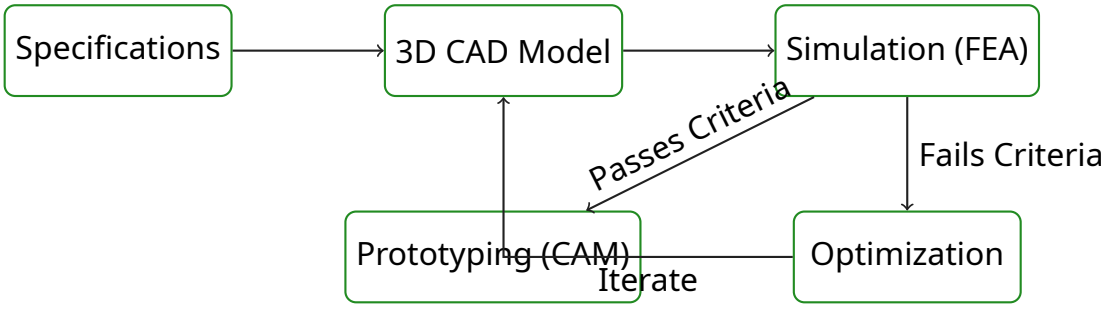


Figure 1.3: Iterative CAD to CAM Engineering Workflow

1.4.4 Advantages

- **High Precision:** Design tolerances can be strictly controlled.
- **Time-Saving:** Accelerates the product development lifecycle.
- **Cost Reduction:** Virtual testing reduces the need for physical prototypes.
- **Easy Modification:** Parametric design allows for rapid iteration and prototyping.